

1. Which of these activities can lead to trash in the ocean?

- A. Throwing a gum wrapper out of a car window
- B. Accidentally leaving sand toys on the beach
- C. Wind blowing trash out of a full trash bin
- D. All of the above

2. Who is in charge of keeping trash out of the ocean?

- A. The President
- B. Beach visitors
- C. Trash collectors
- D. Everyone

3. True or False?

The foam cups used to keep hot drinks hot and cold drinks cold are made out of plastic.

4. Marine debris is...

- A. An animal that lives in the deepest part of the ocean
- B. Any kind of trash that ends up in the ocean
- C. A small plant that floats on the surface of the ocean
- D. The seaweed that washes onto the beach

5. What is the main cause of marine debris in the ocean?

- A. Land-based waste such as littering
- B. Ships dumping trash into the ocean
- C. Plants and animals in the ocean
- D. Tsunamis, hurricanes and tornados (storms) blowing debris into the ocean

6. True or False?

The ocean is always downstream.

7. Trash from which state can end up in the ocean?

- A. Florida
- B. Michigan
- C. Colorado
- D. Every state

8. Which of these is NOT a result of marine debris?

- A. A harmful bloom of algae (red tide)
- B. A seagull getting tangled in fishing line
- C. A person stepping on broken glass on the beach
- D. A turtle eating a plastic bag

9. The "Great Pacific Garbage Patch" is most like...

- A. A floating island of trash you can see from space
- B. A plastic soup
- C. A place for ships to take their trash
- D. All of the above

10. True or False?

Marine debris is too small to damage boats.

TOP 15 MARINE DEBRIS ITEMS

RANK	MARINE DEBRIS ITEM	TOTAL COLLECTED	FUN FACT
1	Cigarette Butts	2,043,470	That's enough cigarettes that when laid end to end, they are the same length as 4,257 school buses. Did you know these butts (the filters) are actually made from type of plastic called cellulose acetate?
2	Food Wrappers	1,685,422	Did you know that most food wrappers, including chip bags and candy wrappers are actually plastic? (See Marine Debris Composition to learn more.)
3	Plastic Beverage Bottles	940,170	That's enough beverage bottles to give every fan attending the Super Bowl 11 sodas.
4	Plastic Bottle Caps	847,972	That is enough bottle caps that when laid end to end, they would cover 3 football fields.
5	Straws/Stirrers	555,007	In the United States alone, 500 million straws are used and thrown away every single day. (www.ecocycle.org/bestrawfree).
6	Plastic Grocery Bags	441,493	That's enough grocery bags that when combined, they weigh more than a pickup truck.
7	Glass Beverage Bottles	394,796	All glass bottles can be beautiful and are great for reusing as vases or for other craft projects.
8	Other Plastic Bags	389,088	These include garbage bags, sandwich bags, clothing store bags, newspaper bags and more.
9	Paper Bags	368,746	It is estimated Americans consume more than 10 billion paper bags each year (www.inteplast.us/ibs/InteGreen/facts.html). That's a lot of bags!
10	Beverage Cans	339,170	Did you know once recycled, an aluminum can becomes a new can in as little as 60 days (www.kab.org/site/PageServer?pagename=recycling_facts_and_stats)?
11	Plastic Lids	312,996	This category includes lids for to-go drinks such as soda and coffee.
12	Metal Bottle Caps	304,638	Metal bottle caps are great for crafts. Collect metals caps to use as decorations for picture frames or games like checkers.
13	Plastic Cups and Plates	282,743	That is enough cups and plates to host a block party for every single person in Newark, New Jersey.
14	Plastic Takeout Containers	234,692	Next time, think about reusing these to-go food containers for storing things such as baseball cards or craft supplies.
15	Other Plastic/Foam Packaging	233,595	This category includes plastic tarps, crates, fishing bait containers and the foam packaging that surrounds new appliances and electronics. While it may seem crazy, even foam is a type of plastic!

Source: 2013 International Coastal Cleanup.

Please visit www.oceanconservancy.org/our-work/international-coastal-cleanup/ for the most up to date Top 10 List or www.coastalcleanupdata.org to find out your local top debris items found.

MARINE DEBRIS SOURCES

OBJECTIVE: Participants will discover the main sources of marine debris.

Define the term *source* with participants.

“ASK: Where do you think marine debris comes from? or What is the source of marine debris?

Marine debris sources are broken into two main categories: ocean-based and land-based. The majority of marine debris originates on land.

1. Ocean-Based Sources:

- **Fishing Vessels:** Fishing gear such as fishing lines and traps can be lost from fishing boats (referred to as derelict fishing gear).
- **Recreational Boaters:** Trash and fishing gear can fall overboard if not stored properly.
- **Stationary Platforms** (Oil and gas drilling platforms): Hard hats, gloves, pipe protectors and 55 gallon drums can all be lost from platforms.
- **Cargo and Other Vessels:** Ships caught in rough seas can lose a variety of items that are being transported, including entire shipping containers. Shipping containers carry many products long distances from where they were made to where they will be sold. Products from sneakers to bath toys have been lost from cargo ships.

2. Land-Based Sources:

- **Litter:** Any trash that is not properly disposed of can end up in waterways and eventually in the ocean.
 - Trash cans or recycling containers may not be readily available, which can lead to inappropriate disposal. Some people simply do not put trash where it belongs. Examples of littering include: leaving food wrappers at the park, throwing fishing line from a boat, or releasing balloons into the sky.
 - Littering can also be accidental. Examples include: a plastic grocery bag flying out of a car window or losing a ball at the beach.
- **Dumping:** Disposal facilities for large or hazardous items may be difficult to find, or they may charge fees. Rather than pay these fees, people may dump large materials such as construction materials, appliances, furniture, mattresses and hazardous waste near creeks or rivers.
- **Storm Water Discharges:** Storm drains carry litter and runoff to waterways that lead to the ocean. Any trash left along the street can easily wash into storm drains and eventually make it to the ocean.
- **Poor Waste Infrastructure:** In some parts of the world, there are no landfills or recycling centers for peoples' trash. In fact, in many places, trash cans do not exist and trash is simply piled on the street or in dry riverbeds. Without a confined place for trash, it is very easy for waste to end up in the ocean.
- **Natural Disasters:** Events such as tornados, hurricanes, floods and tsunamis can scatter debris into the marine environment.



MARINE DEBRIS COMPOSITION

Up to this point, participants have discovered a lot about marine debris including what it is and where it comes from. Next, participants will learn about composition and what happens to trash once it enters waterways.

“ASK: Does anyone know what *compose* or *composition* means?

Allow participants to make guesses, and foster the discussion by providing hints. Then explain the definition of composition to participants.

“ASK: What types of materials are marine debris items composed of?

To encourage responses, **“ASK:** Do you remember the top marine debris items? What materials are they made of (or composed of)?

Marine debris can be composed of a wide variety of different materials, but usually, marine debris is made of one or a combination of the following:

- Plastic
- Glass
- Metal
- Paper
- Cloth
- Rubber
- Wood

“ASK: Which material do you think people find the most of when cleaning up marine debris?

- During the annual International Coastal Cleanup, over 80% of the marine debris items collected by citizen scientists are plastics.
- If time allows, **“ASK:** Why do you think volunteers find so much plastic?
- Plastic is light, and much of it floats, so it's more apt to be collected during cleanups. Also, many of the items we use every day are made of plastic, so there is high plastic consumption and plastic waste.

“SAY: We've learned that a lot of marine debris is composed of plastics, like plastic bottles and fishing line. But did you know that many other items are composed of plastic, too?

“ASK: What about chip bags? Does anyone think those are made of plastic?

Chip bags, even the shiny metallic inside, are composed of thin layers of a type of plastic.

“SAY: Even foam is a form of plastic. Take-out food containers, packing peanuts, foam cups and packaging that protects electronics (or items) like televisions are all a type of plastic.

MARINE DEBRIS DECOMPOSITION

“SAY: Since we understand what *composition* means, can we figure out the definition of *decomposition*?

Once participants have the chance to guess the meaning, explain decomposition and biodegradation.

The formal definition of *decomposition*: the process of breaking down into pieces or simpler elements by natural processes, chemicals, or some other force (Oxford & Merriam-Webster).

The formal definition of *biodegradation*: a process by which microbial organisms transform or alter (through metabolic or enzymatic action) the structure of chemicals introduced into the environment (EPA).

DECOMPOSITION RATES

These decomposition rates are an insightful way for us to look at the long-term impacts of marine debris in the future. Let's revisit plastics.

“ SAY: In the previous activity, we saw that it takes a plastic bottle about 450 years to decompose. But does plastic ever completely return to a natural state?

The answer is no. Plastics are different from the other materials that make up marine debris. Plastics do not biodegrade the way other materials do. Instead, they photodegrade. Explain that this means that instead of being broken down naturally by other organisms, plastics are broken down into smaller and smaller pieces by the sun, wind and waves, but they never completely go away.



ACTIVITY: PLASTIC BREAKDOWN

OBJECTIVE: Participants will understand the difference between the breakdown of plastics and the decomposition of other materials and will see firsthand how photodegradation works.

MATERIALS:

- One piece of paper for each participant. It does not have to be particularly big. The size of your hand will do.

INSTRUCTIONS:

1. Instruct participants to take their pieces of paper and see how many times they can break or rip that piece of paper into smaller and smaller pieces.
2. By the end, participants will likely have a pile of very small pieces that they can no longer rip.

DISCUSSION:

- Explain that this activity mimics how plastics break down in the ocean. They are still there (as your piece of paper is), but they have simply been broken into many smaller pieces.
- Explain that these tiny pieces of plastic are called microplastics. Tiny plastic pieces are especially harmful as they are easily ingested by animals and are very difficult to remove from the marine environment.

PRE-CLEANUP OBJECTIVE 2: Inland to Ocean Concept, Ocean Currents and Gyres

In the previous objective, participants learned about marine debris: what it is, where it comes from and what it is made of. This next section will discuss how marine debris travels from land to sea. And once in the sea, how does it travel across oceans?

TRASH TRAVELS THROUGH WATERSHEDS

To show how trash travels from inland waterways to the ocean, participants first need to understand the concept of a watershed.

“ ASK: How do you think trash travels to the ocean?

- **Trash is carried** from its original resting spot via wind or rain to storm drains.
- **Storm drains carry** trash directly to waterways like streams and rivers.
- **Following the path** of their watershed, those rivers transport the trash to the ocean, resulting in marine debris (EPA).

Gauge participants' understanding of the term *watershed*.

“ ASK: As a group, can we come up with a definition for the word *watershed*?

"A watershed is the area of land where all of the water that falls in it and drains off of it goes into the same place" (USGS). Watersheds come in all shapes and sizes. They cross county, state, and national boundaries. In the continental U.S., there are 2,110 watersheds; including Hawaii, Alaska and Puerto Rico, there are 2,267 watersheds (EPA).

OCEAN SURFACE CURRENTS: THE MARINE DEBRIS HIGHWAYS

Surface ocean currents are mainly driven by global wind patterns. You can think of wind as a solid object that scrapes along the top of the ocean and pulls water in the direction that it's blowing. Ocean currents are made more complex because of land masses, the uneven heating of Earth, and the fact that Earth spins about its axis. (NOAA)

“ SAY: Currents are important because they carry nutrients and organisms (like Crush!) throughout the ocean, sustaining countless marine habitats and wildlife.

Currents are also important because they regulate Earth's climate. The Gulf Stream Current brings warm water from the equator along the east coast of the United States and eventually toward England. This current keeps Northern Europe much warmer than many places as far north.

Currents, both at the surface and deep within the ocean also carry trash.

Circular currents cause the accumulation of marine debris in specific areas.

- **Gyres, or large rotating ocean currents**, can trap trash and marine debris at their centers. This can also happen on a smaller scale as a result of eddies and other factors (NOAA).
- **Gyre currents rotate** clockwise in the Northern Hemisphere and counterclockwise in the Southern Hemisphere. Worldwide, there are five major subtropical oceanic gyres: the North and South Pacific Subtropical Gyres, the North and South Atlantic Subtropical Gyres, and the Indian Ocean Subtropical Gyre. (NOAA)
- **One of the most well-known gyres** is the North Pacific Gyre. This area is also referred to as the “Great Pacific Garbage Patch” and has received a great deal of media attention.

“ ASK: Has anyone ever heard of the “Great Pacific Garbage Patch”?

“ ASK: What do you think this “garbage patch” looks like?

Participants may start to describe a large floating island of trash.

The name “Great Pacific Garbage Patch” has led many to believe that this area is a large and continuous patch of easily visible marine debris, like an island that is visible from space. This is not accurate.

Higher concentrations of trash items can be found in this area, along with other debris such as derelict fishing nets, but most of the debris are actually very small pieces of plastic (NOAA).

The garbage patch is not so much an island as it is a plastic soup. Imagine the garbage as the vegetables and the ocean as the broth.

Just like in soup, the pieces of trash in the gyre collect at different levels in the water column, not just at the surface.

The debris is continuously mixed by wind and wave action and widely dispersed both over huge surface areas and throughout the top portion of the water column (NOAA).

The North Pacific Gyre is well known for the debris that has gathered at its center; however it is not the only plastic soup—marine debris accumulates in every ocean gyre.

OBJECTIVES:

1. To Discuss Reactions to the Cleanup Experience
2. To Relate to the Items Collected and Their Journey
3. To Think Creatively and Critically about Ocean Trash Prevention

REACTING AND RELATING TO CLEANUP

Now that participants have learned all about marine debris and conducted their own cleanup, it is important for individuals to share their reactions to the cleanup experience. This will help conclude the program and move participants to focus on marine debris prevention. Participants now have hands-on experience with the information discussed throughout the Preliminary activities and should be ready to discuss the issue in an action oriented and prevention focused way.

“**ASK** the participants:

- **How did this experience make you feel?**
- **Did you feel frustrated? Surprised? Motivated?**
- **Were you surprised by some of the items found?**
- **Which items surprised you the most? Why?**
- **Were you surprised by the quantity (amount) of certain items?** If you have time to total participants' data forms before the discussion, “**ASK: Were you surprised by the top 5 collected?**
- **How many of the items you collected do you recognize? Do you use many of these items at home? Could any of the items be found in your lunch box?**

- **Bioaccumulation:** the accumulation of a substance (as a pesticide) in a living organism ¹
- **Biodegrade:** capable of being slowly destroyed and broken down into very small parts by natural processes, bacteria, etc. ¹
- **Composition:** the way in which something is put together or arranged; the combination of parts or elements that make up something ¹
- **Decomposition:** the process of breaking down into pieces or simpler elements by natural processes, chemicals, or some other force ¹
- **Eddy:** a circular current of water or air ¹
- **Entangle:** to cause (something) to get caught in or twisted with something else ¹
- **Ghostfishing:** lost or abandoned fishing gear that continues to catch fish ²
- **Gyre:** large system of rotating ocean currents that spiral around a central point ³
- **Ingest:** to take (something, such as food) into your body; to swallow (something) ¹
- **Litter:** Rubbish such as paper, tins, and bottles left lying in an open or public place ⁴
- **Marine Debris:** any persistent solid material that is manufactured or processed and directly or indirectly, intentionally or unintentionally, disposed of or abandoned into the marine environment or the Great Lakes; trash and other solid material that enters oceans and coastal waters ^{3, 5}
- **Microplastics:** pieces of plastic that are less than 5mm long; Microplastics are often created by the breakdown of larger pieces of plastic over time. They can also be manufactured. For example, pre-production industrial plastic resin pellets (nurdles) or plastic “micro-scrubbers” in face wash are considered microplastics. ³
- **Photodegrade:** to be broken down by the action of light, especially sunlight ⁴
- **Prevention:** the act or practice of stopping something from happening or existing ¹
- **Recycle:** to make something new from something that has been used before; to convert waste into a reusable material ^{1, 4}
- **Red tide:** a harmful algal bloom that occurs when colonies of algae grow out of control and discolor water while producing toxic or harmful effects on people, fish, shellfish, marine mammals, and birds ³
- **Reduce:** to make smaller or less in amount, degree or size ⁴
- **Reuse:** to use again especially in a different way or after reclaiming or reprocessing; to use more than once ^{1, 4}
- **Source:** a place, person, or thing from which something comes from or can be obtained ⁴
- **Upcycle:** to reuse discarded objects or material in such a way as to create a product of a higher quality or value than the original, for example, upcycled art or upcycled furniture ⁴
- **Watershed:** the area of land where all of the water that is under it or drains off of it goes into the same place ⁵

SOURCES: ¹ Merriam-Webster; ² Food and Agriculture Organization of the United Nations; ³ NOAA; ⁴ Oxford Dictionaries; ⁵ EPA